
Spacetime Structure and Temporal Dynamics of Mass Deficit Nodes

Curvature Minima, Causal Separation, and the Evolution of Void-Node Geometry

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Abstract

In two companion papers we introduced Mass Deficit Nodes (MDNs) as geometric curvature minima organizing large cosmic voids and demonstrated that photon propagation through MDN-organized spacetime admits a non-canceling, path-integrated redshift contribution consistent with relativistic optics. In this third paper we investigate the spacetime structure and temporal dynamics of MDNs themselves.

We examine the minimal geometric conditions under which MDNs exist as spacetime objects, focusing on curvature minima, causal structure, and time behavior rather than observational magnitude. Using controlled inhomogeneous metric constructions, we show that MDN interiors naturally approach a Λ -dominated geometric ground state, establishing a lower bound on attainable curvature analogous to a zero-point energy. Within this geometry, null and timelike trajectories become categorically separated as a criticality parameter approaches unity, producing horizon-like behavior for null geodesics without singularities or trapped surfaces. We further analyze the role of radiation stress–energy as a self-limiting backreaction and argue that MDNs deepen over cosmic time as radiation density redshifts away.

These results establish MDNs not merely as static features of large-scale structure, but as dynamically evolving curvature-minimum regions with distinctive causal and temporal properties that are geometrically complementary to those of mass-concentration nodes such as black holes. For descriptive clarity, we refer to Mass Deficit Nodes which asymptotically approach a Λ -set geometric floor as void nodes, emphasizing their role as curvature-minimum organizing centers complementary to black holes. No modification of Einstein’s equations is introduced, and no claim of cosmological dominance is made.

1. Introduction

Standard cosmology treats large-scale underdense regions primarily as the absence of matter relative to a mean background. In Paper 1, we proposed a geometric reframing: that the largest and most persistent voids are organized around Mass Deficit Nodes (MDNs) as local minima of spacetime curvature that are geometrically complementary to the curvature maxima associated with mass-concentration nodes such as black holes and dense halos. Under Curvature Extremum

Duality, spacetime is organized by both curvature maxima and curvature minima, each with distinct but related physical consequences.

In Paper 2, we demonstrated that photon propagation through MDN-organized (void-node-organized) geometry admits a non-canceling, path-integrated redshift contribution arising from null–timelike congruence decoupling, fully consistent with relativistic optics and requiring no modification of general relativity.

Having established the conceptual framework and optical consistency of MDNs, the remaining question is spacetime-structural rather than observational:

What kind of spacetime geometry realizes an MDN, and what does that geometry imply for causality, time, and long-term evolution?

This paper addresses that question. We do not attempt to fit observations, revise cosmological parameters, or assert dominance over background expansion. Instead, we identify the minimal geometric, causal, and temporal structures implied by the existence of curvature minima organizing void-dominated regions of spacetime.

2. MDNs as Curvature Minima: Geometric Characterization

2.1 Scalar definition of curvature minima

A Mass Deficit Node is defined geometrically as a local minimum of spacetime curvature. A natural coordinate-independent scalar for identifying such minima is the Kretschmann scalar

$$K \equiv R_{abcd}R^{abcd},$$

which measures total curvature magnitude independent of coordinates or observers.

An MDN location satisfies

$$\nabla_a K = 0, \nabla_a \nabla_b K \text{ positive definite,}$$

evaluated on an appropriate spacelike hypersurface.

This definition is purely geometric. Matter underdensity, void evacuation, suppressed Ricci focusing, and null congruence divergence follow as consequences of the curvature minimum rather than inputs that define it. Observationally inferred density minima act as tracers of MDN-organized regions but do not uniquely define the node itself.

2.2 The geometric floor set by the cosmological constant

In a universe with nonzero cosmological constant, flat Minkowski spacetime is not an attainable dynamical endpoint. The lowest-curvature solution is de Sitter spacetime, with curvature set by Λ .

Two immediate implications follow:

1. Absolute curvature minima are asymptotic rather than reachable.
MDN interiors can deepen but cannot collapse to zero curvature.
2. MDNs possess a natural geometric floor.
The cosmological constant plays a role analogous to a zero-point energy, defining the minimum attainable curvature state.

MDNs are therefore bounded curvature minima, not singularities, pathologies, or inverse black holes.

3. Minimal Metric Constructions for MDNs

To study MDN spacetime structure explicitly, we adopt minimal inhomogeneous metrics sufficient to represent curvature minima while remaining analytically transparent.

3.1 Perturbed FLRW construction

We begin with scalar perturbations of FLRW spacetime in longitudinal gauge:

$$ds^2 = -(1 + 2\Phi(x, t)) dt^2 + a^2(t)(1 - 2\Psi(x, t)) \delta_{ij} dx^i dx^j.$$

We introduce a smooth potential basin

$$\Phi(r) = \Phi_0 f(r/R), f'(0) = 0, f''(0) > 0,$$

which generates a curvature minimum at $r = 0$.

This construction encodes MDNs geometrically without imposing density profiles, mass compensation, Swiss-cheese matching, or spherical symmetry.

3.2 Void interior and de Sitter asymptote

Within sufficiently deep basins, where matter density becomes negligible, the metric asymptotically approaches a Λ -dominated form:

$$ds_{\text{int}}^2 \approx -dt^2 + e^{2H_\Lambda t}(dr^2 + r^2 d\Omega^2), H_\Lambda = \sqrt{\Lambda/3}.$$

This de Sitter-like geometry represents the geometric floor toward which deep MDN interiors evolve, consistent with the bounded curvature argument of Section 2.

4. Null and Timelike Trajectory Separation

4.1 Criticality parameter

We define a dimensionless parameter

$$\chi \equiv \frac{H_{\text{void}} R}{c},$$

which measures the degree to which local void expansion approaches causal limits.

Observed voids today satisfy $\chi \ll 1$, but MDN geometry is characterized by the approach to $\chi \sim 1$.

4.2 Timelike geodesics

Timelike worldlines remain confined to filamentary and wall structures once $\chi \gtrsim 1$. Material objects cannot traverse the void interior because the physical expansion rate exceeds any attainable velocity. Matter evacuation and void persistence follow dynamically from this property.

4.3 Null geodesics and horizon-like behavior

Null geodesics, by contrast, traverse MDN interiors. As $\chi \rightarrow 1$, inward-directed null rays satisfy

$$\frac{dr}{dt} \rightarrow 0.$$

This produces horizon-like behavior for null geodesics, without singularities, trapped surfaces, or event horizons. The causal structure is analogous, but not identical, to black-hole horizons, representing a complementary curvature-minimum regime.

This trajectory separation provides the spacetime-level origin of the path-integrated redshift derived in Paper 2.

5. Time Behavior Near Curvature Minima

5.1 Minimal gravitational time dilation

At curvature minima,

$$g_{tt} \rightarrow -1,$$

implying minimal gravitational time dilation. Clocks in void interiors tick faster relative to clocks in filamentary regions, reflecting reduced coupling between time and curvature rather than an acceleration of time itself.

5.2 Temporal complementarity

In de Sitter-like interiors, conformal time maps an infinite proper-time future to a finite interval. Time does not terminate; instead, it becomes asymptotically decoupled from spatial causality.

This behavior represents the temporal complement of black-hole interiors:

- Black holes: time becomes terminal.
- MDNs: time becomes asymptotically free.

6. Radiation Backreaction and Long-Term Evolution

Photons contribute to the stress–energy tensor and therefore backreact on curvature. A photon traversing an MDN perturbs the curvature minimum slightly, but the same geometric mechanism redshifts the photon:

$$T_{ab}^{\gamma} \propto \omega^2 \text{ decreases along the path.}$$

Consequently:

- backreaction is self-limiting,
- the test-particle approximation improves along null propagation,
- MDN interiors deepen over cosmic time as radiation density redshifts away.

MDNs therefore possess a natural evolutionary arrow toward deeper curvature minima without instability.

7. Observational Circularity and Parameter Absorption

Cosmological parameters are inferred from redshift data under the assumption of purely kinematic interpretation. Any mean MDN contribution is therefore absorbed into fitted parameters.

As a result:

- simple estimates represent lower bounds,
- observable signatures lie in line-of-sight variance, not mean shifts,
- MDN geometry remains compatible with current constraints.

This paper provides the spacetime-structural basis for that circularity without attempting parameter refitting.

8. Discussion: MDNs as Spacetime Objects

The results establish MDNs as:

- genuine geometric structures,
- curvature minima bounded by Λ ,
- dynamically evolving regions,
- possessing horizon-like causal properties for null propagation,
- temporally and causally complementary to mass-concentration nodes.

MDNs are therefore not merely “large voids,” but a distinct spacetime regime within general relativity.

9. Limitations and Future Work

This work does not:

- claim uniqueness of MDN realizations,
- assert cosmological dominance,
- simulate structure formation,
- compute observational fits.

Future work may explore alternative metrics, numerical evolution, and observational tests building on Papers 1 and 2.

10. Conclusion

We have presented a spacetime structural analysis of Mass Deficit Nodes as curvature minima embedded in an expanding universe. Using minimal inhomogeneous constructions, we demonstrated that MDNs naturally exhibit null–timelike trajectory separation, Λ -bounded curvature floors, horizon-like null behavior, and distinctive temporal properties.

Together with Papers 1 and 2, this work completes a logically coherent trilogy: identifying curvature minima as spacetime objects, establishing their optical consequences, and analyzing their spacetime structure. Whether MDNs play a measurable role in cosmology remains an empirical question, but the spacetime geometry required for such a role is now shown to be admissible within classical general relativity.

Appendix A (Illustrative)

An Exact Inhomogeneous Metric Realization of MDN Geometry

A.1 Purpose and Scope

This appendix is illustrative rather than foundational. None of the results or conclusions of the main text depend on the material presented here. Its purpose is to demonstrate that the geometric properties attributed to Mass Deficit Nodes (MDNs) as curvature minima, null–timelike trajectory separation, persistent null congruence divergence, and convergence to a Λ -dominated geometric floor, admit realization within exact solutions of the Einstein field equations.

No claim of uniqueness, generality, stability, or physical realism is implied. The constructions serve as existence proofs only, showing that MDN geometry is not restricted to perturbative reasoning.

A.2 Motivation for an Exact Inhomogeneous Metric

To demonstrate compatibility with exact general relativity, we consider the Szekeres class of solutions, which:

- are fully inhomogeneous and anisotropic,
- impose no spherical symmetry,
- are exact solutions of Einstein’s equations,
- contain FLRW and LTB models as limits.

They provide a technical demonstration that MDN-like geometry can exist without perturbative assumptions.

A.3 The Szekeres Class I Metric

The quasispherical Szekeres Class I metric may be written as

$$ds^2 = -dt^2 + \frac{\left(R'(t, r) - R(t, r) \frac{E'(r, p, q)}{E(r, p, q)}\right)^2}{1 - k(r)} dr^2 + \frac{R^2(t, r)}{E^2(r, p, q)} (dp^2 + dq^2),$$

where $R(t, r)$ is the areal radius, $k(r)$ the curvature profile, and $E(r, p, q)$ the anisotropy function.

A.4 Identifying the MDN

An MDN is identified as a coordinate-independent curvature minimum via the Kretschmann scalar

$$\nabla_a K = 0, \nabla_a \nabla_b K > 0.$$

Choosing $k(r)$ with a smooth minimum at $r = 0$ realizes an MDN as the geometric organizing node of the surrounding void.

A.5 Null Congruence Behavior

In MDN interiors, suppression of the Ricci focusing term allows the null expansion $\theta > 0$ to persist over extended affine intervals, producing path-extensive divergence rather than boundary-localized effects.

A.6 Trajectory Separation

Defining a criticality parameter similar to that in the main text, inward null geodesics approach $dr/dt \rightarrow 0$ while timelike trajectories remain excluded, producing horizon-like behavior without singularities.

A.7 Λ -Dominated Asymptote

As matter density vanishes, the spacetime asymptotically approaches a de Sitter-like geometry, confirming the Λ -set curvature floor.

A.8 Interpretation and Limits

This construction demonstrates:

- compatibility with exact GR,
- persistence of null divergence,
- robustness of null–timelike separation.

It does not establish generality or physical realism.

A.9 Outlook

Exact inhomogeneous realizations motivate numerical evolution, alternative constructions, and deeper causal analysis, which are deferred to future work.